Corrected and conduct of chapter 1

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. 11: the multiplication precedes the addition. *Action:* Rewrite the steps of two calculations with parentheses.
2. $1,\ 8,\ \frac{1}{4}$. *Action:* Build the power chain by dividing by 2.
3. $80 \times 1{,}05 = 84$. *Action:* Connect 5 % to $\frac{5}{100}$ then at the coefficient of 1.05.
4. -2 and 10. *Action:* Replace x in parentheses before calculating.
5. $10 - 3 + 1 = 8$. *Action:* Write the list and find the calculation.
6. $0, 1, 2, 3$. *Action:* Draw a table with two columns, row and value; test range(1,4).

Research activity

Offer A: last 45 kg, sum $\frac{6(20+45)}{2}$ = 195 kg. B: $\frac{20(1-1{,}2^6)}{1-1{,}2} = 198{,}5984$ kg. B exceeds A in total, even if some initial terms are lower. Instrumented calculation allowed here; to argue on cumulation and not on the last term alone.

Immediate applications

1. $2, 3, 6, 11$. The formula $w_n = n^2$ +2 fits: it gives the first term and ($n+1)^2+2-(n^2+2) = 2n+1$.
2. $37 - 17 = 5r$, therefore r = 4 then $u_0 = 17 - 3 \times 4 = 5$.
3. $u_{n+1} - u_n = -4 \times 3^n < 0$: decreasing. v_n alternate 2 and -2: neither increasing nor decreasing.
4. 4 + 8 + 16 + 32 = 60, or $\frac{4(1-2^4)}{1-2} = 60$.
5. u_3 = 6: row 3 for ≤ 6, row 4 for < 6.

Exercises: approach and results

Exercise 1.1 · Take charge of the clues. $u_0 = 0, u_1 = -1, u_4$ = 20. Develop with parentheses: $u_{n+1} = 2(n+1)^2 - 3(n+1) = 2n^2 + n - 1$; difference 4 $n - 1$. The sign changes between rows 0 and 1: the sequence decreases first, then grows from 1.

Exercise 1.2 · Choose the right model. $a_n = 100 + 15n$ and $b_n = 100(1{,}1)^n$, therefore $a_2 = 130$, b_2 = 121. A assumes a constant net addition; B a constant net rate. In reality, rounding and the capacity to accommodate limit these models.

Exercise 1.3 · Find a sequence. $u_3 = q^2 u_1$ so $q^2 = 9$. Positivity requires $q = 3$, where $u_0 = 2$ and $u_5 = 486$. Without this specification, $q = -3$ would also be possible, with $u_0 = -2$ and the same $u_5 = 486$.

Exercise 1.4 · Seat rows. The last row has 18 + 11 × 2 = 40 seats. Sum $\frac{12(18+40)}{2}$ = 348. Check: average 29 seats per row, between 18 and 40.

Exercise 1.5 · Accumulate squares. We get 4 then 9 squares. A integer side square n completed by a line and column requires 2 n + 1 boxes. The formula n^2 has the right first term and $(n+1)^2 - n^2 = 2n + 1$. The generation is the same.

Fiscal year 1.6 · Two refunds. For A: $a_n = 600 - 50n$ for $0 \leq n \leq 12$, then 0. For B: $b_n = \frac{600}{2^n}$. $b_5 = 18{,}75$ and $b_6 = 9{,}375$; strict threshold at row 6. B never gives exactly 0 in this ideal model: do not announce a finished refund.

Exercise 1.7 · Read a list. The first three elements are 2, 5, 14. An initial value and five additions give six elements; indices 0 to 5. $u_5 = 365$.

Exercise 1.8 · Deferred review. $u_7 = 12 + 5(-3) = -3$. Six terms: $\frac{6(12-3)}{2}$ = 27. count the passages, the 6 terms.

CQFD, exercise 9 p. 64, questions a and b. a : $-1, 0, 3, 8$. b : $2, \frac{7}{2}, 4, \frac{17}{4}$. The first row is 0 ; the denominator of b is n + 1 and does not cancel on $\mathbb{N}$.

CQFD, exercise 11 p. 64, all questions. From u_0 = 2, the rule $u_{n+1} = 2u_n - 1$ gives $u_1 = 3$, $u_2 = 5$, $u_3 = 9$, u_4 = 17. The rule uses the previous term, not its rank.

MCQ: answers and reasons

MCQ 1.

1. A. $3 \times 4 + 2 = 14$.

2. C. Three additions: 2 + $3 \times 5 = 17$.

3. D. The coefficient is 1 + $\frac{8}{100}$.

4. B. $9 - 3 + 1 = 7$.

5. C. $\frac{20 \times 21}{2} = 210$.

MCQ 2.

1. D. The difference is $-3 \times 2^n < 0$.

2. A. Four terms; $\frac{1 - 2^4}{1 - 2} = 15$.

3. C. $u_3 = 2$, $u_4 = 1$.

4. B. The terms alternate between two distinct values.

5. D. Eleven rows from 0 to 10.

True or false

1. False: $u_n = -2^n$ is strictly decreasing.

2. False: $u_n = 0$ and $v_n = n(n-1)(n-2)$ coincide with ranks 0.12, but not with rank 3.

3. True: each term is 1 and there are n+1 terms.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.